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A Comprehensive Guide to ELISAs

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1. Introduction

The enzyme-linked immunosorbent assay (ELISA) is a robust and highly selective plate-based assay technique for quantifying the levels of biomolecules such as proteins, peptides, hormones or antibodies, using either an antibody or an antigen as a biorecognition agent. The most crucial element of the detection strategy is a highly specific antibody-antigen interaction. Samples routinely used in ELISAs include serum, plasma, cell culture supernatants, cell lysates, saliva, tissue lysates and urine, as well as food and environmental samples.

ELISAs are typically performed in 96- or 384-well microplates coated with an antibody that will capture a specific target analyte, immobilizing it to the microplate surface to make it easy to separate the bound target from other molecules present. The non-bound materials can then be washed away, before a substrate solution is added to the plate to generate an enzyme-triggered change in color, fluorescence or luminescence proportional to the amount of bound target molecule.

Since their initial development in the 1960s, ELISAs have become a staple in various fields, such as clinical diagnostics, biomedical chemistry, endocrinology, biopharmaceutical analysis, environmental monitoring, security and food testing. Common applications include the detection of various infectious diseases, therapeutic drug monitoring and pregnancy tests.

ELISAs are preferred in many cases due to their sensitivity, specificity, accuracy, reproducibility, and tolerance to harsh buffers or pretreatments. An advantage of ELISAs over other platforms is the ability to customize the assay very specifically to the target analyte or isoform, minimizing interference caused by other biological interactions. ELISAs also offer high throughput, and numerous samples can be analyzed simultaneously, significantly reducing the average analytical time. They are very versatile and can be performed manually or automated.

2. Basic ELISA Protocol

ELISAs all include the same basic steps regardless of the assay variant:

1. **Coating:** A majority of ELISA kits use microplates that have been pre-coated with a target specific antibody/antigen. Alternatively, a coating step is required to coat the plate with the relevant antibody/antigen.
2. **Blocking:** Following antibody binding, a blocking agent (an irrelevant protein or another molecule) is added to the reaction to block all of the unsaturated surface binding sites on the microplate wells.
3. **Analyte capture:** The sample is then added to the plate, and the target molecule is immobilized onto the surface.
4. **Probing/detection:** Following a series of washes to remove non-target molecules that may interfere with the assay, the target molecule is probed by incubation with analyte-specific enzyme-conjugated detection antibodies that affinity-bind to the target to form an antibody-antigen complex. In some assay formats, the detection antibody is not enzyme-conjugated, but acts as a binding site for a secondary conjugate. Enzymes commonly conjugated to detection antibodies include alkaline phosphatase (AP) and horseradish peroxidase (HRP), and a cascade of antibodies can be used to amplify the signal. Following this complex formation step, another series of washes is performed to remove all unbound antibodies, ensuring that only specific (high affinity) binding events are maintained. A substrate is then added to produce a colorimetric signal.
5. **Signal measurement:** Finally, the signal generated by the enzyme-catalyzed reaction is read. The signal observed is proportional to the amount of antigen in the original sample. Various substrates are available for performing the ELISAs with HRP or AP conjugates. The choice of substrate depends upon the required assay sensitivity and the instrumentation available for signal detection, e.g., spectrophotometer, fluorometer or luminometer. It is the detection step that largely determines the sensitivity of an ELISA.

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1. Add 100 µl coating antibody to each well

2. Cover plate, incubate for 1 hour at room temperature to overnight at 4 °C

3. Wash plate three times

4. Add 200-300 µl blocking buffer to each well

5. Cover plate and incubate at room temp for 1 hour to overnight at 4 °C

6. Remove blocker, add sample or standards to each well in duplicate or triplicate

7. Cover plate and incubate at room temperature for 1 hour

8. Wash plate three times

9. Add 100 µl prepared biotnylated detection antibody to each well

10. Cover plate and incubate at room temperature for 1 hour

11. Wash plate three times

12. Add 100 µl prepared enzyme conjugate to each well

13. Cover plate and incubate at room temp for 30 minutes

14. Wash plate 3-4 times

15. Add 100 µl prepared substrate solution to each well

16. Develop at room temperature for 15-30 minutes, before stopping reaction with appropriate stop solution

17. Measure absorbance using appropriate hardware

18. Analyze data and plot signal vs. concentration of antigen

3. Sample Preparation

The amount and type of sample preparation required prior to ELISA processing will vary depending on the sample type, sample origin, assay format and specific study requirements. This section provides generalized examples of sample preparation protocols for various sample types, which can be adapted according to individual needs.

A. Cell lysates

Solubilize the cells in lysis buffer, and allow the sample to sit on ice for 30 minutes. Then centrifuge tubes at 14,000 x g for 5 minutes to remove any insoluble material. Transfer the supernatant into a new tube, discard the remaining whole cell extract, and quantify the total protein concentration in the supernatant using a total protein assay. Assay immediately, or aliquot and store at -20 °C or below.

B. Tissue lysates

Rinse tissue with PBS, cut into 1-2 mm pieces, and homogenize in PBS. Add an equal volume of Radioimmunoprecipitation assay (RIPA) buffer containing protease inhibitors and lyse tissues at room temperature for 30 minutes with gentle agitation. Centrifuge to remove debris. Quantify total protein concentration. Assay immediately or aliquot and store at -20 °C or below.

C. Cell culture supernatants

Centrifuge cell culture media at 1,500 rpm for 10 minutes at 4 °C to remove particulates and assay immediately. Alternatively, aliquot supernatant immediately and store samples at -20 °C or below (preferably -80 °C), avoiding repeated freeze-thaw cycles.

D. Cell extracts

Place tissue culture plates on ice. Remove the media and gently wash cells once with ice-cold PBS. Remove the PBS and add 0.5 ml extraction buffer per 100 mm plate. Tilt the plate and scrape the cells into a pre-chilled tube. Vortex briefly and incubate on ice for 15-30 minutes. Centrifuge at 13,000 rpm for 10 minutes at 4 °C (this creates a pellet from the insoluble content). Aliquot the supernatant into clean, chilled tubes (on ice) and store samples at -80 °C, avoiding freeze/thaw cycles.

E. Conditioned media

Plate the cells in complete growth media (with serum) until the desired confluence is achieved. Remove the growth media and gently wash cells using 2-3 ml of warm PBS. Repeat the wash step. Remove the PBS and gently add serum-free growth media. Incubate for 1-2 days. Remove the media into a centrifuge tube. Centrifuge at 1,500 rpm for 10 minutes at 4 °C. Aliquot the supernatant and keep samples at -80 °C, avoiding freeze/thaw cycles.

F. Serum

Using a serum separator tube (SST), allow samples to clot for 30 minutes at room temperature before centrifugation for 15 minutes at 1000 x g. Remove serum and assay immediately, or aliquot and store samples at -20 °C or below. Avoid repeated freeze-thaw cycles.

G. Plasma

Collect plasma using EDTA, heparin, or citrate as an anticoagulant. Centrifuge for 15 minutes at 1000 x g within 30 minutes of collection. For platelet-poor plasma, a second centrifugation step at 10,000 x g for 10 minutes at 2-8 °C is recommended for complete platelet removal. Assay immediately or aliquot and store samples at -20 °C or below. Avoid repeated freeze-thaw cycles.

H. Urine

Aseptically collect the first urine of the day (mid-stream), directly into a sterile container. Centrifuge to remove particulate matter. Assay immediately, or aliquot and store at -20 °C or below. Avoid repeated freeze-thaw cycles.

I. Saliva

Collect saliva in a tube and centrifuge for 5 minutes at 10,000 x g. Collect the aqueous layer and assay immediately, or aliquot and store samples at -20 °C or below. Avoid repeated freeze-thaw cycles.

J. Milk

Centrifuge for 15 minutes at 1000 x g at 2-8 °C. Collect the aqueous fraction and repeat this process 3 times. Assay immediately.

K. Tissue homogenates

The preparation of tissue homogenates will vary depending upon tissue type. Rinse tissue with 1X PBS to remove excess blood, homogenize in 20 ml of 1X PBS and stored overnight at -20 °C or below. Perform two freeze-thaw cycles to break the cell membranes, then centrifuge for 5 minutes at 5000 x g. Aliquot the supernatant and assay immediately. Alternatively, aliquot and store samples at -20 °C or below. Avoid repeated freeze-thaw cycles.

L. Tissue extracts

Mince tissue on ice in ice-cold buffer, preferably in the presence of protease inhibitors. Place the tissue in microfuge tubes, and dip into liquid nitrogen to snap freeze. Keep samples at -80 °C for later use or keep on ice for immediate homogenization.

For every 5 mg of tissue, add 300 µl of extraction buffer to the tube and homogenize:

- 100 mM Tris, pH 7.4
- 150 mM NaCl
- 1 mM EGTA
- 1 mM EDTA
- 1 % Triton X-100 0.5 %
- 0.5 % sodium deoxycholate

This portion of the buffer can be prepared ahead of time and stored at 4 °C. Immediately before use, the buffer must be supplemented with phosphatase inhibitor cocktail (as directed by manufacturer), protease inhibitor cocktail (as directed by manufacturer) and 1 mM PMSF to generate a complete extraction buffer solution.

Rinse the blade of the homogenizer twice with 300 µl extraction buffer. Shake the sample at 4 °C for 2 hours.

Centrifuge the sample for 20 minutes at 13,000 rpm at 4 °C. Aliquot the supernatant into pre-chilled tubes on ice. Keep the samples at -80 °C, avoiding freeze/thaw cycles.

Note: Lysis buffer volume must be determined according to the amount of tissue present. Typical concentration of final protein extract is at least 1 mg/ml.

4. ELISA Controls

ELISA controls verify that your assay was run properly and that everything is performing accurately. The most basic control is the blank sample control, but additional controls are needed to provide a comparison to real life conditions, and to ensure that the assay continues to provide accurate results. Control samples that have had their analyte concentration validated by another method should be used.

A. Positive controls

A positive ELISA control can be a recombinant or natural sample that you know will definitely be detectable in the assay. Positive controls serve to show that a negative sample is truly negative. A standard curve is a form of positive control.

B. Spiked controls

It is important to ensure that there is nothing present in the sample matrix that interferes with assay performance when using complex samples. Spike controls can indicate assay performance by calculating the percent recovery from the ELISA protocol. Spiking a known amount of recombinant or natural analyte into your sample matrix allows you to verify that the readout matches the expected value. When recombinant proteins are used, their equivalent functionality to the endogenous wild type versions of the protein needs to be checked.

C. Negative controls

Negative controls help to check that you are not obtaining any false positive results or non-specific binding. To do this, use a sample that you know does not express or contain the protein you are measuring. For example, if you are quantifying a cell culture supernatant, a good negative control would be to test your cell culture media. False positive ELISA signals can be identified by assaying the linearity of dilution.

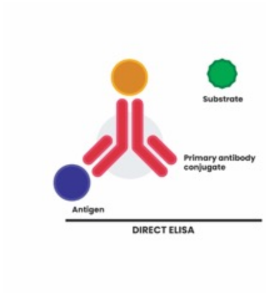
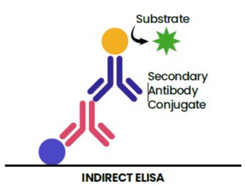
D. Standards

A standard contains a known concentration of the target protein from which the quantification curve can be obtained. A poor standard curve may indicate that the antibody didn't bind properly, or that it doesn't capture the protein in the standard.

5. ELISA Protocol

ELISAs can be grouped into four main categories – direct, indirect, sandwich, and competitive – based on how the analytes and antibodies interact. Each type of ELISA has its own advantages and disadvantages, as illustrated in Table 1 below.

Table 1: Advantages and disadvantages of different types of ELISA.

Format	Advantages	Disadvantages
	• Entails fewer steps, resulting in a quicker protocol. • Less prone to processing errors, as fewer steps and reagents are required. • Cross-reactivity of secondary antibodies is eliminated.	• Antibody immobilization is not specific, so may cause higher background noise than the indirect ELISAs, as all the proteins in the sample will bind to the plate, not just the target. • Immunoreactivity of primary antibody might be adversely affected by labeling with enzymes or tags. • Labeling primary antibodies for each specific ELISA system is time-consuming and expensive. • No flexibility in choice of primary antibody label from one experiment to another. • Minimal signal amplification. • Low sensitivity.
	• Highly sensitive, as more than one labeled secondary antibody can bind the primary antibody. • Economical, as fewer labeled antibodies are needed. • Very flexible since different primary antibodies can be used with a single labeled secondary antibody. • A wide variety of labeled secondary antibodies are available commercially. • Maximum immunoreactivity of the primary antibody is retained because it is not labeled.	• Risk of higher background noise, as the secondary antibody may be cross-reactive, resulting in non-specific signal. • Longer protocol than the direct ELISA, due to additional incubation step for secondary antibody.

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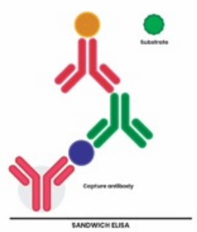
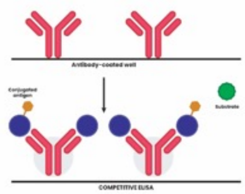
<table>
<tr><th>Format</th><th>Advantages</th><th>Disadvantages</th></tr>
<tr><td>
</td><td>More sensitive than the direct or indirect ELISAs.
Highly specific, as two antibodies are involved in capture and detection, and the antigen/analyte is specifically captured and detected.
Very flexible, since both direct and indirect detection can be used.
Suitable for complex (or crude/impure) samples, minimizing sample purification requirements.</td><td>Antibody optimization can be difficult because cross-reactivity may occur between the capture and detection antibodies, requiring a standardized ELISA kit or tested antibody pair.
Longer protocol.
Challenging and time-consuming to develop.</td></tr>
<tr><td>
</td><td>Little or no sample preparation is required, and crude or impure samples can be used.
More robust than sandwich ELISAs, as less sensitive to sample dilution and sample matrix effects.
More consistent, due to less variability between duplicate samples and assays.
Extremely flexible because it can be based on direct, indirect or sandwich ELISAs.
Able to quantify small molecules that cannot easily be bound with two antibodies.
Rapid protocol.</td><td>Same limitations as the base protocol it is adapted from.
Low specificity, as only uses one antibody.
Requires a conjugated antigen.</td></tr>
</table>

Recommended materials will vary slightly based on which ELISA protocol is best suited for the end-user. The preparation of the following ELISA buffers remains the same regardless of which protocol is chosen.

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Materials

- Antigen (preferably purified)/Standards/Controls
- Antibodies required for the assay
- Coating buffer: 0.05 M carbonate-bicarbonate, pH 9.6
- Wash solution: 50 mM Tris, 0.14 M NaCl, 0.05 % Tween® 20, pH 8.0
- Blocking (post-coat) solution: 50 mM Tris, 0.14 M NaCl, 1 % BSA, pH 8.0
- Sample/Conjugate diluent: 50 mM Tris, 0.14 M NaCl, 1 % BSA, 0.05 % Tween 20, pH 8.0
- Enzyme substrate: TMB
- Stop solution: 0.18 M H2SO4 (or other appropriate solution)
- High protein binding microplate (e.g., Thermo Scientific Clear C-Shaped Immuno Nonsterile 96-Well Plates, #446612)

Preparation

Bethyl Laboratories' ELISA Starter Accessory Kit (E101) or ELISA Starter Accessory Kit II (E103) may be used, or you can prepare reagents as specified below. The E103 kit contains non-BSA buffers for ELISAs where BSA or other bovine, goat or sheep proteins are being detected.

Coating Buffer
- 3.7 g sodium bicarbonate ($NaHCO_3$)
- 0.64 g sodium carbonate ($Na2CO_3$)
- 1 l distilled water

Tris buffered saline (TBS)
- 6.06 g Tris base
- 8.2 g NaCl
- 6.0 ml 6 M HCl
- 1 l distilled water
- pH should be 7.2 to 7.8, conductivity should be 14,000 to 16,000

Wash solution
- 1 l of TBS
- 5 ml 10 % Tween 20

Blocking solution
- 1 l of TBS
- 10 g BSA

Conjugate diluent
- 1 l TBS
- 10 g BSA
- 5 ml 10 % Tween 20

0.18 M H_2SO_4
- Stock is 18 M
- 10 ml stock solution
- 1 l distilled water

A. Direct ELISA Protocol

In the direct ELISA method, the antigen is directly immobilized to the surface of the well of an ELISA plate, then detected with an antigen-specific enzyme-conjugated antibody. Substrate is then added, producing a signal that is proportional to the amount of analyte in the sample. HRP and AP substrates typically produce a colorimetric output that is read by a spectrophotometer. Different concentrations of the sample should be prepared in coating buffer, and identical volumes added directly to the plate. This technique is best for analyzing the immune response to an antigen.

The below example protocol outlines the procedure for performing a direct ELISA method.

Procedure:

Perform all steps at room temperature.

Coat with antigen

1. Dilute the antigen in coating buffer to the desired concentration. For purified antigens, 1-2 µg/ml is sufficient. For non-purified antigens, optimal concentration needs to be determined. Transfer 100 µl of diluted antigen to each well.
2. Incubate coated plate for 60 minutes.
3. After incubation, aspirate the antigen solution from each well.
4. Wash each well with wash solution as follows:
    - i. Fill each well with wash solution.
    - ii. Remove wash solution by aspiration.
    - iii. Repeat for a total of 3 washes.

Blocking

1. Add 200 µl of blocking solution to each well.
2. Incubate 30 minutes.
3. After incubation, remove the blocking solution and wash each well three times.

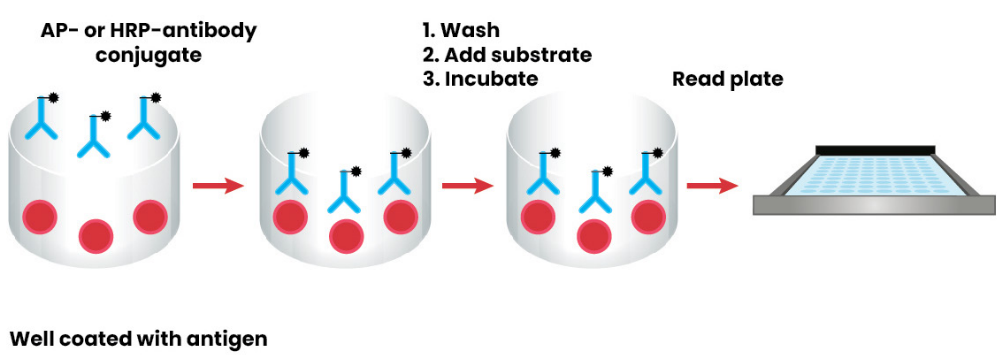
HRP-conjugated primary antibody

1. Dilute the HRP conjugate in conjugate diluent.
2. Transfer 100 µl to each well.
3. Incubate 60 minutes.
4. After incubation, remove HRP conjugate and wash each well 5 times.

Enzyme substrate reaction

1. Prepare the substrate solution according to the manufacturer's recommendation. TMB is highly recommended, but OPD or ABTS can be used.
2. Transfer 100 µl of substrate solution to each well.
3. Incubate plate 5-30 minutes.
4. To stop the TMB reaction, apply 100 µl of 0.18 M H_2SO_4 to each well. If using another substrate, use the stop solution recommended by manufacturer.
5. Using a microplate reader, read the plate at the appropriate wavelength for the substrate (450 nm for TMB).

Notes: This procedure is suitable for AP-conjugated antibody. Use an appropriate substrate for AP. OD = optical density.



B. Indirect ELISA Protocol

The indirect ELISA method is similar to the direct method, but instead an unlabeled and unconjugated antigen-specific primary antibody binds to the antigen, then a secondary enzyme-labeled antibody binds to the primary antibody. A substrate is then added, producing a signal proportional to the amount of secondary antibody present. The indirect ELISA is best for determining the total antibody concentration in samples.

The example protocol below outlines the procedure for performing an indirect ELISA.

Procedure:

Perform all steps at room temperature.

Coat with capture antibody

1. Determine the number of single wells needed. Standards, samples, blanks and/or controls should be analyzed in duplicate.
2. Dilute the antigen in coating buffer to the desired concentration. For purified antigens, 1-2 μg/ml is sufficient. For non-purified antigens, optimal concentration needs to be determined. Transfer 100 μl of diluted antigen to each well.
3. Incubate coated plate for 60 minutes.
4. After incubation, aspirate the antigen solution from each well.
5. Wash each well with wash solution as follows:
    - i. Fill each well with wash solution.
    - ii. Remove wash solution by aspiration.
    - iii. Repeat for a total of 3 washes.

Blocking

1. Add 200 μl of blocking solution to each well.
2. Incubate 30 minutes.
3. After incubation, remove the blocking solution and wash each well three times.

Primary antibody

1. Dilute the HRP-conjugated secondary antibody in conjugate diluent to the optimal concentration.
2. Transfer 100 μl to each well.
3. Incubate 60 minutes.
4. After incubation, remove HRP-conjugated secondary antibody solution and wash each well 5 times.

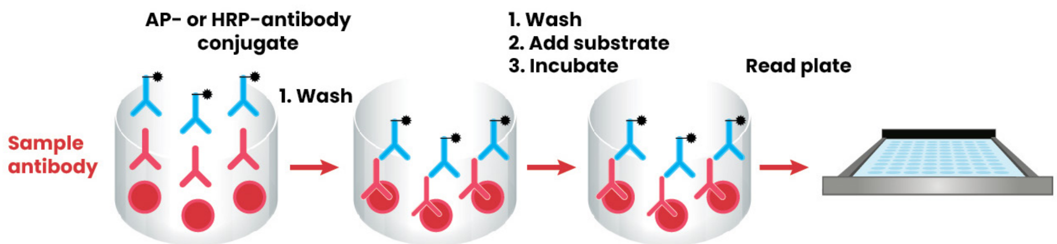
HRP-conjugated secondary antibody

1. Dilute the HRP-conjugated secondary antibody in conjugate diluent to the optimal concentration.
2. Transfer 100 μl to each well.
3. Incubate 60 minutes.
4. After incubation, remove HRP-conjugated secondary antibody solution and wash each well 5 times.

Enzyme substrate reaction

1. Prepare the substrate solution according to the manufacturer's recommendation. TMB is highly recommended, but OD or ABTS can be used.
2. Transfer 100 µl of substrate solution to each well.
3. Incubate plate 5-30 minutes.
4. To stop the TMB reaction, apply 100 µl of 0.18 M H_2SO_4 to each well. If using another substrate, use the stop solution recommended by the manufacturer.
5. Using a microplate reader, read the plate at the appropriate wavelength for the substrate (450 nm for TMB).

Notes: This procedure is suitable for AP-conjugated antibodies. Use an appropriate substrate for AP.



C. Sandwich ELISA

The sandwich ELISA method is different from the previous two methods in that the surface of the plate is first coated with a target-specific capture antibody before the sample is added. If any of the target antigen is present in the sample, it binds to the capture antibody. A detection antibody is then added, which also binds to the antigen, before an enzyme-linked secondary antibody is added, which binds to the detection antibody. Once a substrate is added, a signal proportional to the amount of analyte present in the sample is produced. Sandwich ELISAs typically require the use of matched antibody pairs, where each antibody is specific for a different, non-overlapping part (epitope) of the antigen molecule. This makes them highly specific and very sensitive, which is why sandwich ELISAs are the most common type of ELISA. They are best for the analysis of complex samples, since the antigen does not need to be purified prior to measurement.

The example protocol below outlines the generalized method for a sandwich ELISA, as well as troubleshooting and technical information to ensure your method is a success.

Procedure:

Perform all steps at room temperature.

Coat with antigen

1. Determine the number of single wells needed. Standards, samples, blanks and/or controls should be analyzed in duplicate.
2. Dilute capture antibody to a concentration of 2-10 µg/ml in coating buffer. Transfer 100 µl to each well.
3. Incubate coated plate for 60 minutes.
4. After incubation, aspirate the capture antibody solution from each well.
5. Wash each well with wash solution as follows:
   i. Fill each well with wash solution.
   ii. Remove wash solution by aspiration.
   iii. Repeat for a total of 3 washes.

Blocking

1. Add 200 µl of blocking solution to each well.
2. Incubate 30 minutes.
3. After incubation, remove the blocking solution and wash each well 3 times.

Standards and samples

1. Dilute the standards in sample diluent according to desired concentration.
2. Dilute the samples, based on the expected concentration of the analyte, to fall within the concentration range of the standards.
3. Transfer 100 µl of standard or sample to assigned wells.
4. Incubate plate 60 minutes.
5. After incubation, remove samples and standards and wash each well 3 times.

HRP-conjugated primary antibody

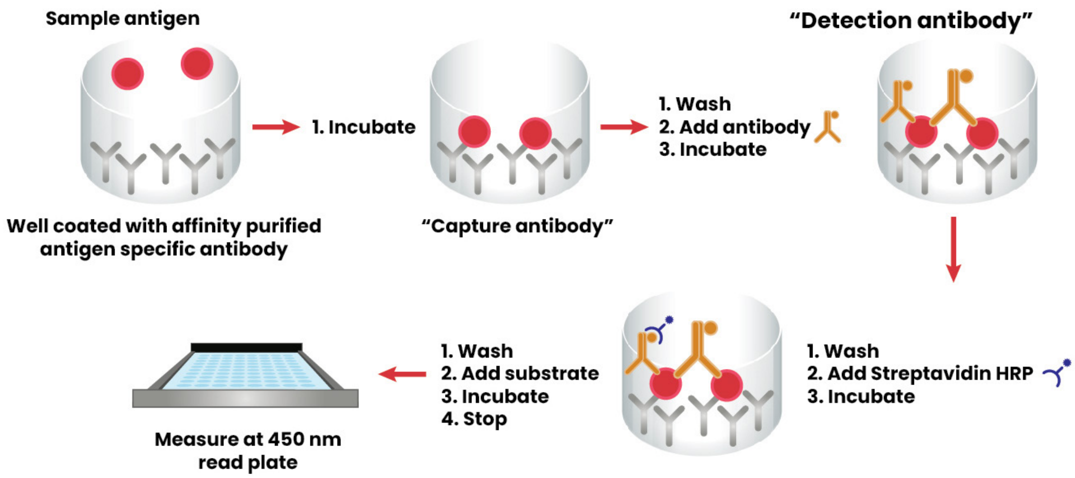
1. Dilute the HRP-conjugated primary antibody in conjugate diluent.
2. Transfer 100 µl to each well.
3. Incubate 60 minutes.
4. After incubation, remove HRP-conjugated antibody solution and wash each well 5 times.

Enzyme substrate reaction

1. Prepare the substrate solution according to the manufacturer's recommendation. TMB is highly recommended, but OD or ABTS can be used.
2. Transfer 100 µl of substrate solution to each well.
3. Incubate plate 5-30 minutes.
4. To stop the TMB reaction, apply 100 µl of 0.18 M H_2SO_4 to each well. If using another substrate, use the stop solution recommended by the manufacturer.
5. Using a microplate reader, read the plate at the appropriate wavelength for the substrate (450 nm for TMB).

Calculation of results

1. Average the duplicate readings from each standard, control, and sample.
2. Subtract the zero reading from each averaged value above.
3. Create a standard curve by reducing the data using computer software capable of generating a four-parameter logistic curve fit. Other curve fits may also be used.
4. A standard curve should be generated for each set of samples.

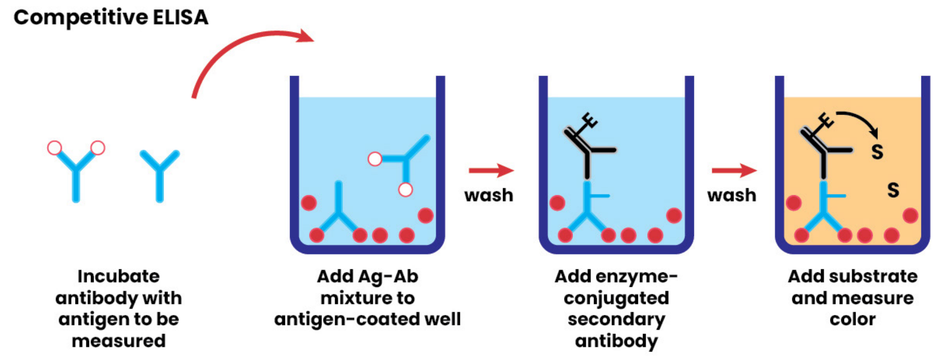


D. Competitive ELISA protocol

A competitive ELISA, also known as the 'labeled analyte' technique, involves a competitive reaction between antigens present in the sample and antigens bound to the wells of a microplate to a specific amount of labeled antibody. The reference antigen is pre-coated on a multi-well plate and the sample is pre-incubated with labeled antibody, and then added to the wells. Depending on the amount of antigen in the sample, more or less free antibodies will be available to bind the reference antigen. Some competitive ELISAs use labeled antigens instead of labeled antibodies, where the labeled antigen and the sample antigen compete for binding to the unlabeled primary antibody. A substrate is then added, and the signal produced is inversely proportional to the amount of antigen present in the original sample. Amounts of labeled and non-labeled antigen to use in the assay should be determined empirically. Competitive ELISAs are commonly used for determining the concentrations of small molecules and hormones, when the analyte of interest is too small to efficiently sandwich with two antibodies. Each of the previously mentioned ELISA formats can be adapted to the competitive format.

The example protocol below outlines a competitive ELISA based on an indirect method.

1. Coat microplate wells with 100 µl of the antigen solution, at a concentration of 1-10 µg/ml in coating buffer. Cover the plate and incubate overnight at 4 °C. Wash the plate 3 times using wash buffer.
2. Add 200-300 µl of blocking solution to each well. Incubate for 3 hour at RT. Wash 3 times using wash buffer.
3. Prepare the antigen antibody mixture by adding 50 µl of antigen to 50 µl of antibody for each well in the assay (use a range of antigen concentrations appropriately diluted in diluent). Incubate for 1 hour at RT.
4. Add 100 µl of the above mixture to each well. Incubate for 1 hour at RT. Wash 3 times using wash buffer.
5. Add 100 µl enzyme-conjugated secondary antibody (appropriately diluted in diluent) to each well. Incubate for 1 hour at RT. Wash 3 times using wash buffer.
6. Add 100 µl of the appropriate substrate solution to each well. Incubate at room temperature (and in the dark if required) for 15-30 minutes, or until desired color change is attained.
7. To stop the reaction, apply 100 µl of appropriate stop solution to each well.
8. Read absorbance values immediately at the appropriate wavelength. Gently tap plate to ensure thorough mixing. Measure absorbance within 30 minutes.



6. Troubleshooting Your ELISA

Low or absent signal

Possible Causes	Solutions
Incorrect dilutions or pipetting errors may cause the components to be at a low concentration	Determine optimal dilution of each component by a titration assay.
One of the reagents may have degraded or become contaminated	Replace the reagents with fresh and uncontaminated reagents.
Incorrect reagents added	Check protocol, ensure correct reagents are added in proper order, and prepared to correct concentrations.
The antibodies do not bind effectively to the antigen or work in combination with each other	Test alternative antibodies.
Insufficient or no antibody	Increase concentration of antibodies to find optimum.
Capture antibody (or antigen) does not bind to plate/plate not coated properly	Use ELISA plate, not tissue culture plate. Try longer coating time and different coating buffers. Increase concentration of coating components. Try avidin plates with biotinylated antigen.
The detection system (substrate) may not be sensitive enough, or target is present below the detection limit	It may be necessary to either concentrate the sample, decrease dilution factor or switch to a more sensitive substrate.
Blocking protein in coating solution	Eliminate blocking protein from coating solution
Incubation time too short	Follow the manufacturer guidelines and, if the problem continues, incubate the samples at 4 °C overnight.
Incubation temperature too low	Ensure incubations are done at correct temperature. Before proceeding, all reagents should be at room temperature or per manufacturer guidelines.
Incompatible sample type	Use sample types that the assay is known to detect; include positive controls in your experiment.
Incompatible assay buffer	Ensure assay buffer is compatible with the target of interest.

Possible Causes	Solutions
Enzyme inhibitor present	Avoid sodium azide in HRP reactions. Avoid phosphate in AP reactions.
Incorrect storage of components	Double check storage conditions on assay kit; most kits need to be stored at 4 °C.
Very vigorous plate washing	Gently pipette wash buffer (manual method). Ensure correct pressures (automatic wash system).
Wells dry out	Cover using plate sealer for all incubations.
Wells scratched with pipette or pipette tips	Carefully dispense and aspirate solutions into and out of wells.
Plate read at incorrect detection wavelength/ wrong filter in microplate reader	Use recommended wavelength/filter; wavelength should be 450 nm for TMB, 490 nm for OPD, or 405 nm for ABTS Ensure plate reader is set up correctly for substrate used.
Slow color development	Prepare substrate immediately before use. Allow longer incubation, at correct temperature, ensuring plates and reagents are kept at room temperature beforehand. Ensure stock solution hasn't expired, is uncontaminated and is maintained at correct pH. Increase antibody concentration and substrate quantity.
Epitope recognition impeded by adsorption to plate	Conjugate peptide to large carrier protein before coating onto plate or coat at a higher concentration.

Saturated Signal

Possible Causes	Solutions
High sample concentration	Dilute samples further, determining the optimal dilution by a titration assay.
Excessive substrate	Decrease concentration or amount of substrate, follow manufacturer guidelines.
Substrate color changed before use	Make substrate immediately before use. Never use substrate that has expired or exhibits a color change when poured into reservoir.
Non-specific antibody binding	Try different formulations of coating solutions. Ensure the plate is suitable for use in ELISA applications. Use affinity-purified antibody, preferably one that is pre-adsorbed. Use serum (5-10 %) from same species as secondary antibody for the blocking buffer.
Incubation time too long	Follow the manufacturer guidelines, or apply stop solution earlier than recommended.

Possible Causes	Solutions
Excess antibody	Decrease antibody concentration to find optimum.
Buffers cross-contaminated with metals, HRP, other samples or positive control	Make fresh buffers in clean, unused containers.
Insufficient washing	Follow the manufacturer guidelines. If problem persists, add an extra wash step. At the end of the washing step, gently pat dry on a clean paper towel.
Plate sealers not used or re-used	During incubations, cover plates with plate sealers. Use a fresh sealer every time.
Plate read at incorrect detection wavelength/ wrong filter in microplate reader	Use recommended wavelength/filter; wavelength should be 450 nm for TMB, 490 nm for OPD, or 405 nm for ABTS Ensure plate reader is set-up correctly for substrate used.
Excess time before plate reading	Read your plate within 30 minutes after adding the substrate and, if you cannot perform the reading within this timeframe, add a stop solution after sufficient color is developed in the plate to prevent further progress.

High background levels

Possible Causes	Solutions
Incorrect quantities or concentrations of enzyme conjugate, substrate, antibodies or blocking buffer	Optimize quantities and concentrations of the enzyme conjugate, substrate, antibodies and blocking solution to your experiment. Increase blocking time. Use fresh buffer.
Sample components and antibodies cross-reacting with the blocking buffer	Run appropriate controls. Wash off blocking buffer before adding antibody.
Non-specific antibody binding	Try different formulations of coating solutions. Ensure the plate is suitable for use in ELISA applications. Use affinity-purified antibody, preferably one that is pre-adsorbed. Use serum (5-10 %) from same species as secondary antibody for the blocking buffer.
Too much detection reagent	Dilute reagent.
Contaminated blocking buffer or enzyme substrate	Use fresh blocking buffer and enzyme substrate. Use a clean container and assess color, e.g. TMB should be clear and colorless.
Blanks contaminated with samples	Change pipette tips when switching between blanks and samples. Seal plates to avoid any cross-contamination between wells.
Sample contaminated with enzymes	Test samples with substrate to check for contamination.

Possible Causes	Solutions
Reaction not stopped	Use stop solution to prevent overdevelopment.
Plate left too long before reading	Read your plate within 30 minutes after adding the stop solution.
Wrong settings on plate reader	Use recommended wavelength/filter. Ensure plate reader is set-up correctly for substrate used.
Insufficient Tween 20	Optimize the concentration of Tween 20 in the blocking and wash buffers for your experiment, starting at 0.05 %.
Incubation with substrate carried out in light	Perform incubation in the dark.
Incubation temperature too high	Try different temperatures to find the right one for your assay.
Plates are stacked in incubation, causing uneven temperatures and evaporation rates	Avoid stacking plates. Use plate sealers during incubation periods.
Pipetting errors	Calibrate pipettes. Use proper technique.
Insufficient reagent mixing	Before pipetting, ensure all reagents and sample have been mixed thoroughly.
Inconsistent or insufficient well and plate washing causes interference from dirt or water	Standardize washing procedure. Remove all liquid from each well between washing steps. Use a plate washer that is routinely decontaminated and maintained. Increase number of washing cycles.
Suboptimal salt concentration in washing buffer	Optimize salt concentration; a high concentration can reduce the non-specific interactions.

Low Sensitivity

Possible Causes	Solutions
Assay format not sensitive enough	Switch to a more sensitive detection system, e.g., from colorimetric to chemiluminescence. Switch to a more sensitive assay type, e.g., sandwich ELISA. Increase incubation times and/or temperatures.
Improper storage of ELISA kit	Store all reagents as recommended.
Insufficient target	Reduce sample dilution or concentrate sample.
Inactive substrate	Ensure reporter enzyme has the expected activity.
Poor target adsorption to wells	Covalently link target to wells.
Insufficient substrate	Increase concentration or amount of substrate.
Incompatible sample type	Use sample types that the assay is known to detect; include positive controls in your experiment.

Possible Causes	Solutions
Interfering ingredients in buffers and sample	Check reagents for any interfering chemicals, e.g., sodium azide or EDTA.
Mixing or substituting reagents from different kits	Avoid mixing components from different kits.
Incorrect plate reading setting	Use recommended wavelength/filter. Ensure plate reader is set-up correctly for substrate used.

Signal fades quickly

Possible Causes	Solutions
An excess of HRP can exhaust the substrate, preventing the emission of light	Check that the concentration of the enzyme conjugate is correct for the chemiluminescent substrate used in the assay. It may also be necessary to optimize the antibody concentrations, blocking solution and wash steps.

Fluorescent signal is low

Possible Causes	Solutions
Overexposure of fluorescent reagents to intense light causes photo-bleaching/decomposition of the fluorophore	Protect the fluorophore from strong light.
Fluorophores in close proximity to one another can quench the signal of neighboring molecules through the dispersion of energy	Re-label or use a soluble chemifluorescent substrate.

Slow color development

Possible Causes	Solutions
Substrates too old, contaminated or used at incorrect pH	Make and use fresh substrates at correct pH immediately before use.
Expired/contaminated solutions	Make and use fresh reagents.
Incorrect incubation temperature	Ensure plates and reagents are kept at room temperature before pipetting into wells. During incubation, seal the plate completely with a plate sealer and avoid stacking plates.
Low antibody concentration	Repeat the assay with higher antibody concentrations to find the optimal one for your experiment.

Substrate forms a precipitate or plate glows in the dark

Possible Causes	Solutions
There is too much enzyme present	Dilute the enzyme conjugate, optimize antibody concentrations and blocking, perform additional washing cycles.

Poor standard curve linearity and dynamic range

Possible Causes	Solutions
Causes of a poor dynamic range are the same as for low or absent signal and high background levels If the curve has good linearity but high standard error, there might be inconsistent pipetting between samples or individual users.	All samples and standards should be measured at least in duplicate or triplicate. Pipettes should be calibrated so that they always deliver the same volume from assay to assay.
Improper standard solution	Confirm dilutions are done correctly. Consistently follow the ELISA protocol. Make new standard curve as appropriate.
Standard improperly reconstituted	Briefly spin vial before opening. Inspect for undissolved material after reconstitution.
Standard degraded	Store and handle standard as recommended. Prepare standards no more than two hours before use.
Improper curve fitting	Plot standard data using a different curve fitting model.
Pipetting error	Use calibrated pipettes and proper pipetting technique.
Insufficient washing	Follow the manufacturer guidelines. At the end of each washing step, gently pat dry on a clean paper towel.
Poorly mixed reagents	Thoroughly mix reagents.
Poor/variable adsorption of reagents to plate	Extend incubation time. Check coating buffer. Use a different plate as appropriate. Check homogeneity of sample.
Plates stacked during incubation	Avoid stacking plates. Use plate sealer instead.
Dirty or defective plates	Clean the plate bottom.

Poor replicate data and inconsistent results

Possible Causes	Solutions
Bubbles in wells	Ensure there are no air bubbles prior to reading plate.
Insufficient washing of wells	Carefully wash wells. Follow recommended protocols. Check that all ports of the plate washer are unobstructed.
Incomplete reagent mixing	Ensure all reagents are mixed thoroughly.
Inconsistent pipetting	Use calibrated pipettes and proper pipetting techniques. If a multi-channel pipette is used, ensure that all channels deliver the same volume.
Inconsistent sample preparation or storage during protocol or incubation	Ensure consistent sample preparation and optimal sample storage conditions. Adhere to recommended incubation temperatures. Use the same protocol from run to run.
Particulates in samples	Remove the particulates by centrifugation.
Plate sealers not used or re-used	During incubations, cover plates with plate sealers. Use a fresh sealer every time.
Cross-well and buffer contamination	Ensure plate sealers and pipette tips are not contaminated with reagents. Use fresh buffers.
Edge effects (higher or lower OD in peripheral wells than in central wells)	Ensure plates and reagents are kept at room temperature before pipetting into wells, unless otherwise instructed. During incubation, seal the plate completely with a plate sealer and avoid stacking plates.

7. Optimizing Your ELISA

- Make sure all reagents are brought to room temperature before using (unless instructed to keep them cold).
- If you are not going to run the entire plate, ensure that the remaining strips are sealed in the bag with the desiccant to prevent moisture causing degradation.
- For standards that are not single use, it is best to aliquot the remaining standard into smaller volumes and freeze. This allows you to avoid repeated freeze-thaws.
- Multichannel pipettes speed up plating of standard, samples and reagents, leading to more consistent results.
- When pipetting, dispense liquid with the pipette tips held at an angle, and not touching the sides or bottom of the well.
- While it is not necessary to change your pipette tips between each replicate, it is recommended that you change them between different samples or standards to prevent contamination.
- It is highly recommended that a plate washer is used, as manual plate washing can lead to higher background signals. However, make sure the plate washer has routine preventive maintenance and decontamination performed.
- When washing plates, either manually or with a plate washer, be sure to give the wash buffer time to work by adding a 30 second soak time in between washes.
- Pay close attention to incubation times. As a general guide, the incubation time should not vary by more than ±5 minutes per hour of incubation time.
- Incubation time of the enzyme substrate will depend on the substrate used, and the intensity of the color change. The high standard should have an OD reading of ~2.0-2.5, and the OD reading of the low standard should be above background.
- If the assay calls for incubation in a cold environment – 2-8 °C – and you are running multiple assays, do not stack the plates on top of each other. Instead, place them individually on the shelf.
- Coated, covered plates are stable overnight at 4 °C.
- When preparing the coating buffer from the gel capsule, break the capsule apart and pour ingredients into water. Do not place gel capsule into water. The gelatin from the capsule interferes with the binding of the coating antibody to the plate.
- Capture antibody diluted with coating buffer should be added to wells immediately.
- Check all buffers for contamination and expiration. When troubleshooting, it may be helpful to start with all new buffers. Make buffers in new or properly cleaned vessels.
- Sodium azide should not be added to any of the buffers.
- Dilutions should be made shortly before application and immediately applied to appropriate wells.
- Excess antibody/analyte should be wiped from pipette tips when making dilutions.
- Use caution when pipetting small volumes of antibody/analyte. A minimum of 5 µl is recommended.
- Stop solution should be added to the plate in the same order as the enzyme substrate.

8. ELISA Data

ELISAs can yield three different types of data output:

1. Quantitative

Data can be interpreted in comparison to a standard curve (a serial dilution of a known analyte) to precisely calculate the concentrations of antigen in various samples.

2. Qualitative

Data can also be used to achieve a yes or no answer indicating whether a particular analyte is present in a sample, as compared to a blank well containing no analyte or an unrelated control. The inclusion of a single fixed level positive control can be used to give a yes/no criteria for minimum analyte level.

3. Semi-quantitative

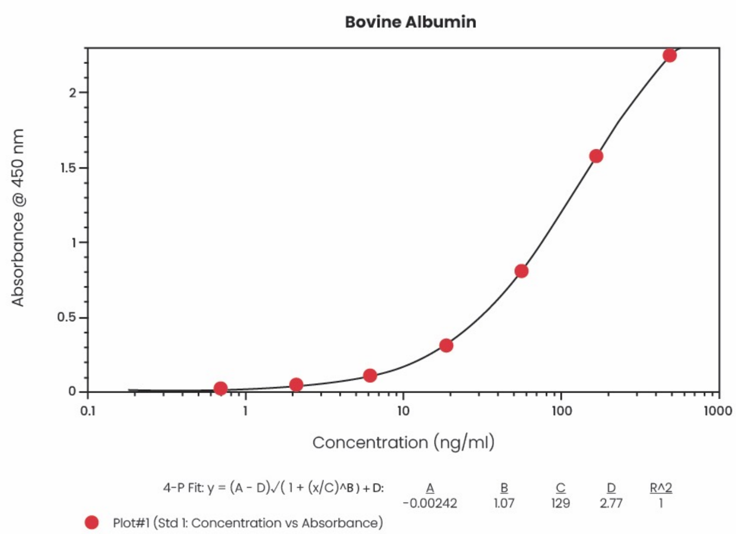
ELISAs can be used to compare the relative levels of analyte in assay samples, since the intensity of signal will vary directly with analyte concentration.

Standard curve and calibration curve models

ELISA data is typically plotted with OD vs log concentration to produce a sigmoidal curve. To calculate a quantitative result, average the duplicate or triplicate readings for each standard, control and sample, then subtract the average zero standard OD. Use this to produce a standard curve, by reducing the data using computer software capable of plotting the mean absorbance (y axis) against the standard concentration (x axis), and then find the line of best fit. If software is unavailable, the data may be linearized by plotting the log of the concentrations versus the log of the OD on a linear scale. The best fit line can be determined by regression analysis. This procedure will produce an adequate, but less precise, fit of the data.

To determine the concentration of an unknown sample, find the absorbance value on the y-axis and extend a horizontal line to the standard curve. At the point of intersection, extend a vertical line to the x-axis and read the corresponding concentration. If the samples have been diluted, the concentration read from the standard curve must be multiplied by the dilution factor.

The relatively long linear region of the curve makes ELISA results accurate and reproducible. This calculation can be done directly on the graph, or you can use curve-fitting software to plot the data using semi-log, log/log, log/logit and the 4- or 5-parameter logistic models. Using a software package with linear regression enables you to check the R^2 value, to assess the overall fit. For the linear portion of the curve, R^2 values >0.99 represent a very good fit. The linear plot compresses the data points on the lower concentrations of the standard curve, making it the most accurate range, i.e., the most likely to achieve the required R^2 value. To counteract this compression, a semi-log chart can be used, where the log of the concentration value (on the x axis) is plotted against the readout. This method gives an S-shaped curve that distributes more of the data points into the sigmoidal pattern. A log/log (log of concentration against log of readout) creates a more linear data curve.



Normalizing data

It is extremely important for ELISAs to be highly precise, so that results are accurate and reproducible from one experiment to the next. Intra-assay precision is the reproducibility between wells within an assay and allows the researcher to run multiple replicates of the same sample on one plate and obtain similar results. Inter-assay precision is the reproducibility between assays and guarantees that the results obtained using multiple kits over a period of time will be reproducible. To limit the impact of these factors on assay results, normalization should be performed, by comparing the standards on each plate between experiments. Precision is measured as a coefficient of variation (CV), the ratio of the standard deviation to the mean, which is usually expressed as a percentage.

9. ELISA Reagents

A. Buffers

Several different buffers are used during an ELISA protocol for coating, blocking, washing, and sample and antibody dilution.

Coating buffers

Coating buffers stabilize the antigen or antibody which is used to coat the ELISA plate, in order to enhance adsorption to the plate and/or detectability. No other proteins should be present in the coating buffer, as they will compete with the antibody/antigen for binding to the plate, reducing the downstream assay signal. The two most common coating buffers are bicarbonate buffer at pH 9.6 or PBS.

Typically, 50-100 µl of coating buffer containing an antigen or antibody at a concentration of 1-10 µg/ml will be added to the ELISA plate and incubated or overnight at 4 °C or for 1-3 hours at room temperature. Try different temperatures, timings, buffers and coating agent concentrations to optimize your individual experiment. It is important to maintain a moist environment in the well during the coating stage to prevent evaporation, for example by using plate sealers.

Our ELISA coating buffer can be purchased at www.fortislife.com/products/elisa-accessories/elisa-coating-buffer/E107.

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Blocking buffers

The blocking stage prevents the detection antibodies from binding directly to the assay plate. Proteins permanently block binding, so are added after the capture antibody has bound to the plate surface, whereas detergents only block temporarily, and are lost during washing. If a protein is used in the blocking buffer, then a detergent should not be added before or during the blocking step, as it will hinder blocking. Basic blocking buffers contain 1 % BSA or milk proteins dissolved in PBS or TBS, and 200-300 µl is added to each well and incubated for 1-3 hours at room temperature.

Our ELISA blocking buffer can be purchased at www.fortislife.com/products/elisa-accessories/elisa-blocking-buffer/E104.

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Washing buffers

Plates should be washed 3-5 times between steps with PBS or TBS containing a nonionic detergent (such as Tween 20) to remove any unbound materials. Excess washing buffer must be completely removed in the final washing step to prevent diluting subsequently added reagents, but bear in mind that the plate should not dry out at all. Plate washers offer the most effective methods of washing plates in between steps.

Our TBS Washing Solution can be purchased at www.fortislife.com/products/elisa-accessories/elisa-wash-solution/E106.

Our 10 % Tween 20 can be purchased at www.fortislife.com/products/elisa-accessories/10-Tween 20/E108.

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B. Standard Curve Dilutions

Dilution accuracy

It is critical that the same units are used for concentrations, volumes, and amounts. Do not mix units when calculating dilutions, as this will cause inaccuracies. Some commonly converted units used in ELISA are:

- 1 liter = 1,000 ml
- 1 ml = 1,000 µl
- 1 g = 1,000 mg
- 1 mg = 1,000 µg
- 1 µg = 1,000 ng

Dilution will always reduce the concentration of the sample. Dilutions are generally expressed as ratios in terms of whole numbers, reduced to the lowest common denominator. The dilution ratio can therefore be defined as the volume of sample per total volume.

Example:

To dilute the human IgG control for a Human IgG ELISA standard curve, you will need to know:

- The standard curve range
- How much Human IgG is in the human IgG control

For this example, the standard curve range is 7.8-500 ng/ml, and the human IgG control contains 4.4 mg/ml of IgG.

First, convert the units as described above, e.g.:

4.4 mg/ml X 1,000 X 1,000 = 4,400,000 ng/ml

The upper limit of the standard curve is 500 ng/ml. Calculate the control dilution required to reach this value, e.g.:

4,400,000 ng/ml ÷ 500 ng/ml = 1/8,800

The first dilution is therefore a 1/8,800 dilution:

1 µl + 8.8 ml

Volumes can be adjusted up or down (e.g., 2 µl + 17.6 ml) to avoid pipetting only 1 µl, as this is less accurate.

For the rest of the standard curve, make 2-fold serial dilutions until you reach 7.8 ng/ml, as below:

500ng/ml	1/8800	2 µl + 17.6 ml
250ng/ml	1/17,600	1 ml + 1 ml
125 ng/ml	1/35,200	1 ml + 1 ml
62.5 ng/ml	1/70,400	1 ml + 1 ml
31.25 ng/ml	1/140,800	1 ml + 1 ml
15.6 ng/ml	1/281,600	1 ml + 1 ml
7.8 ng/ml	1/563,200	1 ml + 1 ml

Common sample dilutions

Some common sample dilutions used in ELISAs are 1/100, 1/1,000, 1/5k, etc. See below for dilution schemes. This list is not meant to be exhaustive but is merely a starting point.

1/50	20 µl sample into 980 µl diluent
1/100	10 µl sample into 990 µl diluent
1/1000	5 µl sample into 4,995 µl diluent
1/10k	Make initial 1/100 dilution (see above), then place 10 µl of it into 990 µl diluent for final dilution of 1/10k
1/50k	make initial 1/1,000 dilution (see above) then place 100 µl of it into 4.9 ml diluent for final dilution of 1/50k
1/100k	make initial 1/1,000 dilution (see above) then place 100 µl of it into 9.9 ml diluent for final dilution of 1/100k

C. Antibodies

Antibodies can be monoclonal or polyclonal. Monoclonal antibodies are homogeneous, and so are specific to a single epitope or region of an antigen. They therefore do not react with other closely related antigens or trigger non-specific signals. Monoclonal antibodies are commonly used as matched pairs in sandwich ELISAs but can be used for capture or detection in combination with a polyclonal antibody to make antigen capture more likely. Matched pairs are sets of antibodies that bind different epitopes on the same antigen, and so are used together to capture and detect a single antigen in a sandwich assay.

Polyclonal antibodies are heterogeneous, so they bind numerous epitopes on a single antigen molecule. They are less likely to not bind, and therefore generate higher detection signals. However, polyclonal antibodies can often produce a higher non-specific signal, due to their broad specificity. Polyclonal antibodies can be conjugated to HRP or AP and can function as both capture and detection antibodies.

The interactions between antibodies and antigens are described in terms of their specificity, affinity and avidity. Specificity describes whether an antibody binds uniquely to one particular epitope on a single type of antigen, or whether it binds to multiple epitopes that are found on numerous different antigens. If the antibody is not specific, it is known as cross-reactive. Affinity denotes the degree, strength and durability of a bond between an antibody and an antigen. High-affinity antibodies bind quickly to a large quantity of antigen, and the bond lasts for a long time, producing many stable connections and a very sensitive detection. Avidity denotes the stability of the antibody-antigen relationship, based upon antibody valence, spatial arrangement, and the total number of antigen binding sites.

Commercially available antibodies are often provided unconjugated, and so you will need to do this yourself. The antibody must first be purified and should be in a 10-50 mM amine-free buffer at a pH of 6.5-8.5. It can then be conjugated with either HRP or AP. It is critical to purify antibodies before conjugation, as they may contain salts, proteins or amines as stabilizers or preservatives, which will affect conjugation.

Bethyl offers both primary- and secondary-conjugated antibodies for use in ELISAs. The full selection of antibodies can be purchased at www.fortislife.com/products/antibodies.

D. Substrate

The last stage of an ELISA is to add an enzyme substrate to the wells, causing an enzyme-catalyzed reaction. This reaction generates a colored, fluorescent or luminescent end product that can be read in a suitable spectrophotometer. For HRP-catalyzed reactions, the unreacted substrate should be colorless or very light yellow in appearance, and when it reacts with peroxidase, it produces a soluble blue color.

Our TMB One Component HRP Microwell Substrate can be purchased at www.fortislife.com/products/elisa-accessories/tmb-one-component-substrate/E102.

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E. Stop Solution

At the end of the substrate incubation period, a volume of ELISA stop solution equal to the volume of substrate solution is added to the wells to halt the ongoing enzymatic reaction. For peroxidase-based detection systems, this results in the rapid color change from blue to yellow. Halting the reaction during the linear phase enables OD readings to be used to calculate the analyte concentration of the unknown samples. The yellow end product is stable for up to 30 minutes after stop solution has been added, so OD should be measured within that time. For the TMB stop solution, 0.18M sulfuric acid can be used.

Our ELISA Stop Solution can be purchased at www.fortislife.com/products/elisa-accessories/elisa-stop-solution/E115. ELISA Stop Solution stops the TMB enzyme substrate reaction.

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10. Our Products

All our ELISA kits are for research use only.

Bovine Albumin ELISA Kit (E11-113)

Reactivity	Bovine
Assay range	0.69-500 ng/ml
Sample types	Serum, plasma, colostrum, milk, cell culture supernatant
Detection method	Biotinylated streptavidin-HRP
Assay type	Sandwich ELISA

Citations

Proliferative regulation of alveolar epithelial type 2 progenitor cells by human Scnn1d gene. Zhao, Ali, Chang et al. Theranostics (2019) 9 (26), 8155-8170 DOI: 10.7150/thno.37023

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Inhibitory effect and cross-link breaking activity of Moringa oleifera leaf crude extracts on fructose-derived advanced glycation end-products. Adeniran, Mogale. South African Journal of Botany (2021) 139, 122-129 DOI: 10.1016/j.sajb.2021.02.006

The sequential role of Mst1/mTORC1/STAT1 activity in chemokine receptor 2-regulated B cell receptor signaling. Zhu, Gu, Yang et al. Authorea (2021)

Human IgM ELISA Kit (E88-100)

Target	Human IgM
Reactivity	Human
Assay range	1.03-750 ng/ml
Sample types	Serum, plasma, cerebrospinal fluid (CSF), milk, urine
Detection method	Biotinylated streptavidin-HRP
Assay type	Sandwich ELISA

Citations

Measuring the Impact of Targeting FcRn-Mediated IgG Recycling on Donor-Specific Alloantibodies in a Sensitized NHP Model. Manook, Flores, Schmitz et al. Front Immunol (2021) 12, 660900 DOI: 10.3389/fimmu.2021.660900

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Epigenetic targeting of Waldenström macroglobulinemia cells with BET inhibitors synergizes with BCL2 or histone deacetylase inhibition. Matissek, Han, Karbalivand et al. Epigenomics (2021) 13 (2), 129-144 DOI: 10.2217/epi-2020-0189

Human IgA ELISA Kit (E88-102)

Target	Human IgA
Reactivity	Human
Assay range	1.03-750 ng/ml
Sample types	Serum, plasma, cerebrospinal fluid (CSF), milk, urine
Detection method	Biotinylated streptavidin-HRP
Assay type	Sandwich ELISA

Citations

Acetate differentially regulates IgA reactivity to commensal bacteria. Takeuchi, Miyauchi, Kanaya et al Nature (2021) 595 (7868), 560-564 DOI: 10.1038/s41586-021-03727-5

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Evaluation of IgA1 O-glycosylation in Henoch-Schönlein Purpura Nephritis Using Mass Spectrometry

Nakazawa, Imamura, Kawamura et al. Transplant Proc (2019) 51 (5), 1481-1487 DOI: 10.1016/j.transproceed.2019.01.122

Human IgG ELISA Kit (E88-104)

Target	Human IgG
Reactivity	Human
Assay range	0.69-500 ng/ml
Sample type	Serum, plasma, cerebrospinal fluid (CSF), milk, urine
Detection method	Biotinylated streptavidin-HRP
Assay type	Sandwich ELISA

Citations

Glutathione peroxidase 4-regulated neutrophil ferroptosis induces systemic autoimmunity. Li, Jiang, Li et al. Nat Immunol (2021) 22 (9), 1107-1117 DOI: 10.1038/s41590-021-00993-3

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Human Albumin ELISA Kit (E88-129)

Target	Human albumin
Reactivity	Human
Assay range	0.69-500 ng/ml
Sample types	Serum, plasma, cerebrospinal fluid (CSF), milk, urine
Detection method	Biotinylated streptavidin-HRP
Assay type	Sandwich ELISA

Citations

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SARS-CoV-2 IgG ELISA Kit (E88-301)

Target	SARS-CoV-2 IgG
Reactivity	Human
Sample types	Serum, plasma
Detection method	HRP
Assay type	Indirect ELISA

SARS-CoV-2 IgM ELISA Kit (E88-302)

Target	SARS-CoV-2 IgM
Reactivity	Human
Sample type	Serum, plasma
Detection method	HRP
Assay type	Indirect ELISA

Mouse Albumin ELISA Kit (E99-134)

Target	Mouse albumin
Reactivity	Mouse
Assay range	1.23-900 ng/ml
Sample types	Serum, plasma, urine
Detection method	Biotinylated streptavidin-HRP
Assay type	Sandwich ELISA

Citations

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11. Frequently Asked Questions

How should I store my samples before use?

We recommend that you store samples at 4 °C. However, biological or cell culture samples can be initially frozen until use but should not be repeatedly frozen and thawed. Once the sample is thawed, it should be stored at 4 °C.

How do I dilute my samples?

Since all experimental designs are different, it is difficult to determine an overall dilution scheme for all samples. If you have a general idea of the concentration of your samples, you will want to make dilutions so that your samples will fall in the middle (linear) range of the standard curve. Otherwise, you may want to take a few of your samples and make several dilutions (e.g., 1:100, 1:1000, and 1:10,000) to determine a dilution range for your samples. Each investigator must determine the best dilution for their samples.

What if my raw OD values are too low or too high?

If the raw OD values appear too low, allow the reaction to continue by incubating for longer than the 30 minutes during the TMB development step, as outlined in our ELISA kit protocol. If the raw OD values appear to be developing too high, apply a stop solution to the wells sooner than the 30-minute TMB incubation time.

What can I do to decrease my background?

Usually, background is caused by cross contamination of samples or reagents. Make sure to always use a plate washer or wash by hand with a multichannel pipette. Do not use the 'squirt and dump' method for washing. You may also add another wash cycle to each step. Make new reagents if background problems continue, and make sure that you are not using a BSA product with an ELISA kit that may cross react (for example, Bovine Albumin ELISA, Bovine Transferrin ELISA, Bovine IgG ELISA and Goat IgG ELISA).

The quantification of my sample is not consistent or seems to be incorrect. What can I do to correct this problem?

Make sure that you are quantifying your samples in the linear region of the standard curve. Samples that quantitate too close to the extreme ends of the curve may be inaccurate and difficult to reproduce.

What kind of software is needed to graph a 4-parameter curve?

Some of the software choices are SoftMax Pro by Molecular Devices, KC Jr., SigmaPlot, or others. You cannot directly plot a 4-parameter curve with Microsoft Excel®.

I do not have software that will perform a 4-parameter standard curve. What should I use to analyze my data?

You can use a linear regression curve in Microsoft Excel. If you use this type of curve, only use a maximum of 5 points on the curve. We recommend that you discard the upper and lower points and plot your standard curve with the 5 middle points.

Why am I seeing a high variability between my sample replicates?

The main reason for high sample variability in an assay is inconsistent pipetting and washing, so it is important to perfect these techniques. However, some of this variability is unavoidable – this is the rationale for calculating the average results from sample replicates. Another possible culprit for high variability is the 'edge effect', where the outermost wells of the plate are more vulnerable to drying out due to evaporation. Plate stacking will also cause variability because temperature will be unevenly distributed across the plates.

Can I extend the standard curve?

No one can guarantee the assay accuracy at concentrations outside of the specified range of the curve are used. A specific range is generated to provide the statistical confidence in the assay's accuracy.

Why do my wells turn green after I add the stop solution?

The green color is a result of incomplete mixing between the substrate and stop solution. After adding the stop solution, gently tap the plate or place it on a shaker until the mixture in the wells turns yellow.

Why does a brown or orange-brown precipitate appear in my wells after adding the stop solution? How can I resolve this issue?

The precipitate is a result from insufficient washing after incubation with the HRP-labeled detection antibody. To resolve this issue, perform a 30-second soak during each wash step followed by a complete removal of all liquid in the wells.

If I don't use all the wells from a microplate for my current ELISA, how can I preserve the unused wells for future use?

ELISA microplates typically have removable strips of wells. Unused wells may be removed from the plate, returned to the foil pouch containing the desiccant pack, and stored at 2-8 °C for up to one month.